

Hands-on Lab: Generative AI for Data Generation and Augmentation

Estimated time needed: **30** minutes

One of the principle advantages of generative AI is its ability to generate realistic synthetic data. The synthetic data is generated when a pretrained generative model responds to either a prompt, create new data samples, or transfers learns on a given data set. In addition, it creates samples that can augment the existing data set while maintaining the statistical distribution and interpretability of the data set.

In this lab, you will learn how to use generative AI to generate synthetic data samples and transfer learns on a given data set.

Learning Objective

In this lab, you will learn how to use a popular tool, [Mostly.ai](#), to create synthetic data samples to augment a CSV data set.

Data Set

You will use a data set that includes insurance records.

The data set is available at the following link:

[Insurance Dataset](#)

This data set is a cleaned-up version of the [Medical Insurance Price Prediction](#) data set, available under the [CC0 1.0 Universal License](#) on the [Kaggle](#) website.

Steps

1. Download the data set

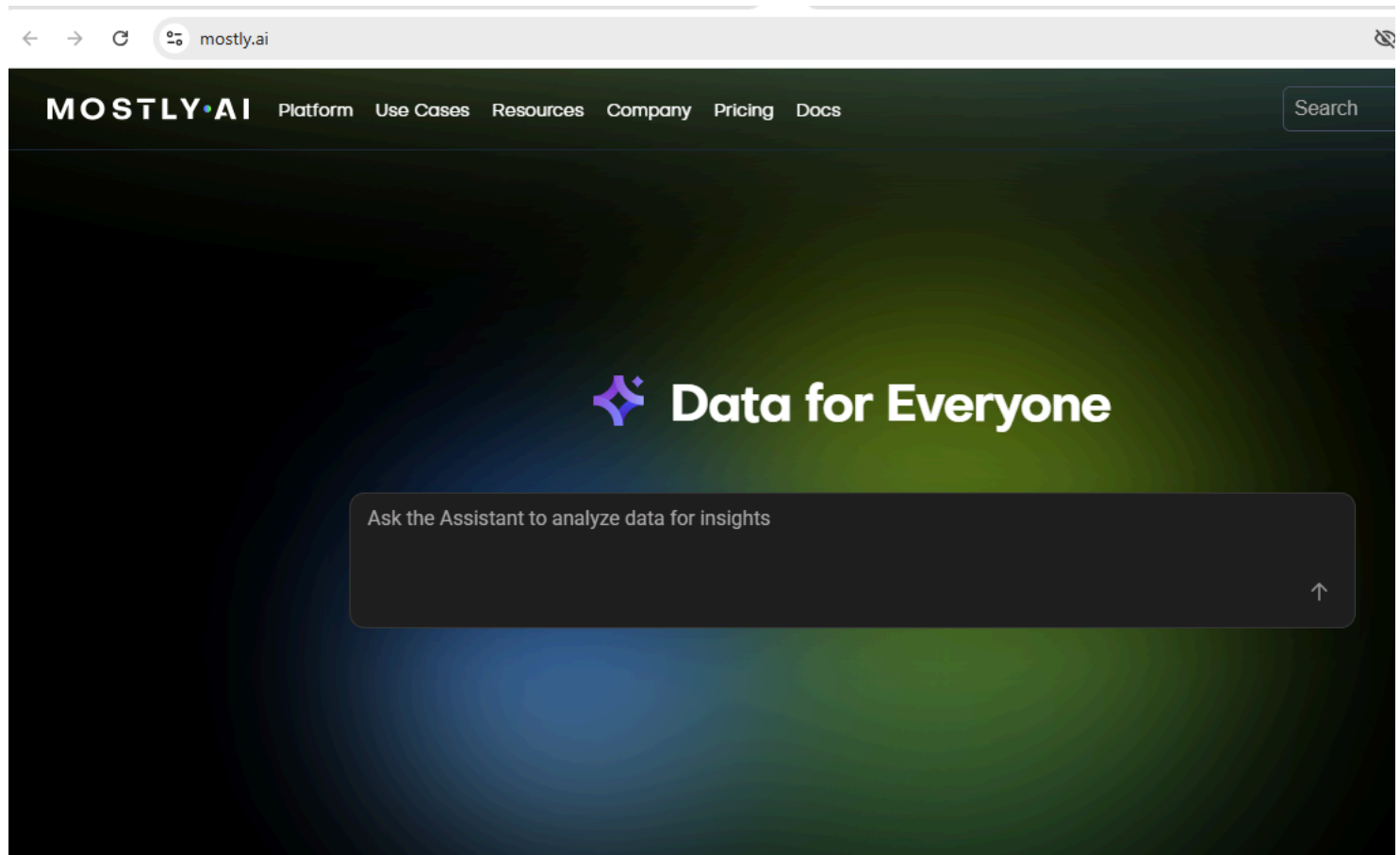
The first step is to download the dataset on your machine. You will need to upload this file to the interface in a subsequent step. Click the link provided in the **Data Set** section to download the data set.

2. Open the website

Click the following link to open the mostly.ai website and interface.

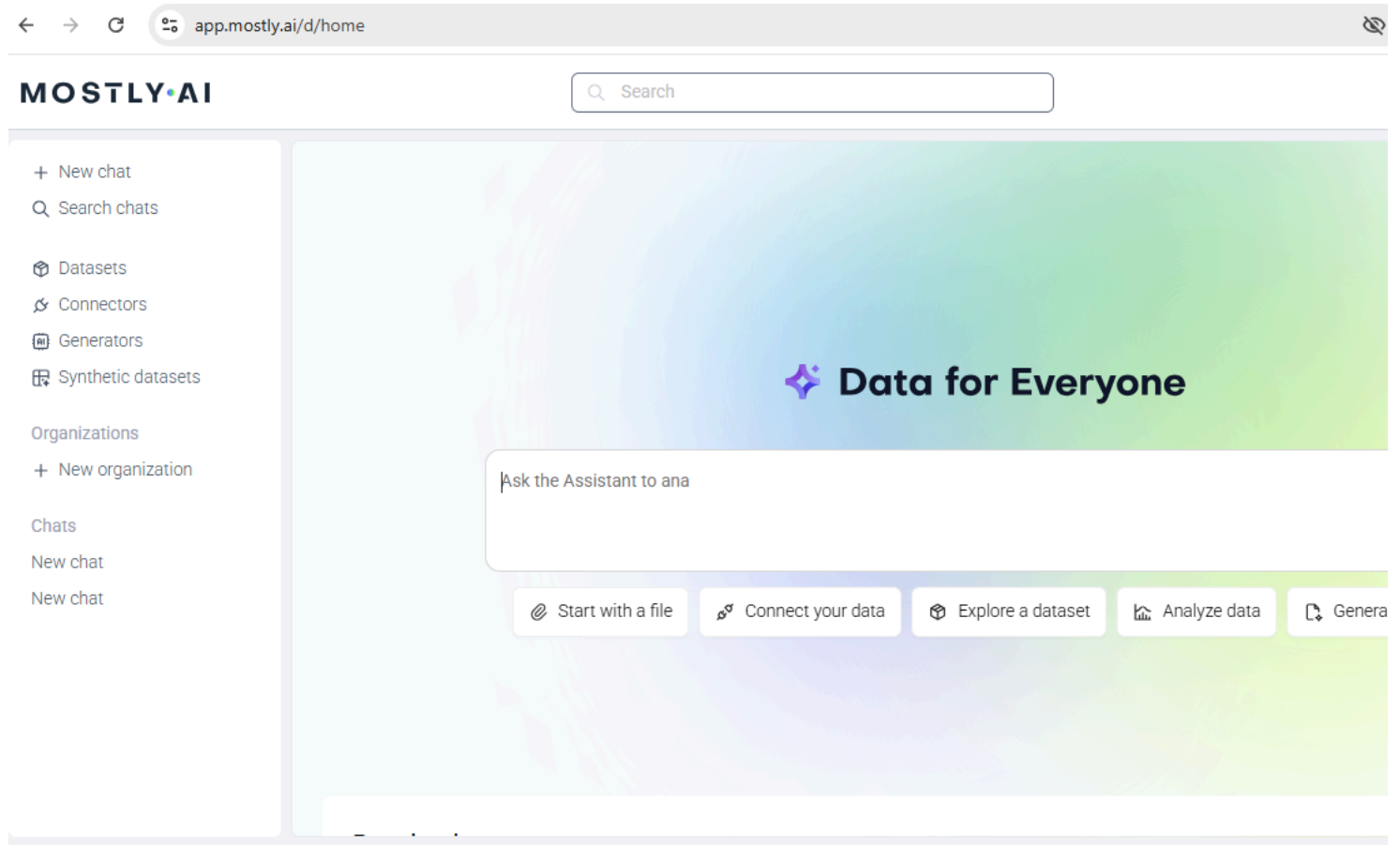
<https://mostly.ai/>

This link opens in a new browser tab, and you should see an web page that looks similar to the following screen capture:



3. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface.



4. Upload the data set

- Click on the **Generators** given on the left hand side of the page.

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- Search chats
- Datasets
- Connectors
- Generators**
- Synthetic datasets

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Data for Everyone

Ask the Assistant to g

- Start with a file
- Connect your data
- Explore a dataset
- Analyze data
- Generators

Popular datasets

- And upload the CSV file of the data set to the interface by using the Upload your data option available on the console.

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Generators

Generators are models that learn from original data. Once trained, they allow to create any number of synthetic samples as well as simulate data.

Train a generator

On platform

Train a generator with your data on platform.

- Start from a connector
- Upload your data**

Locally

Train a generator on your environment and import it to the platform.

- Use the SDK
- Import a generator

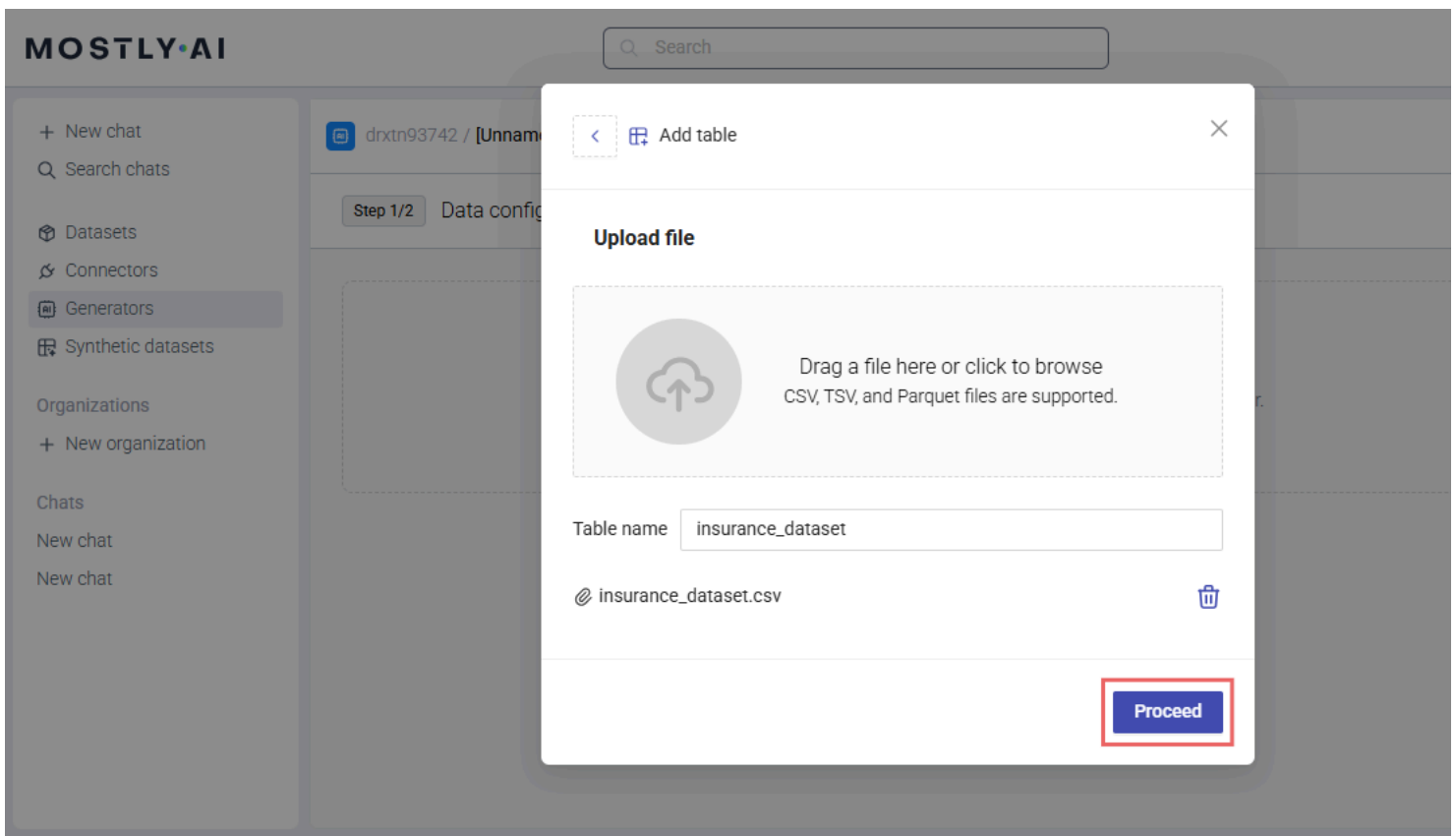
Available generators

Search



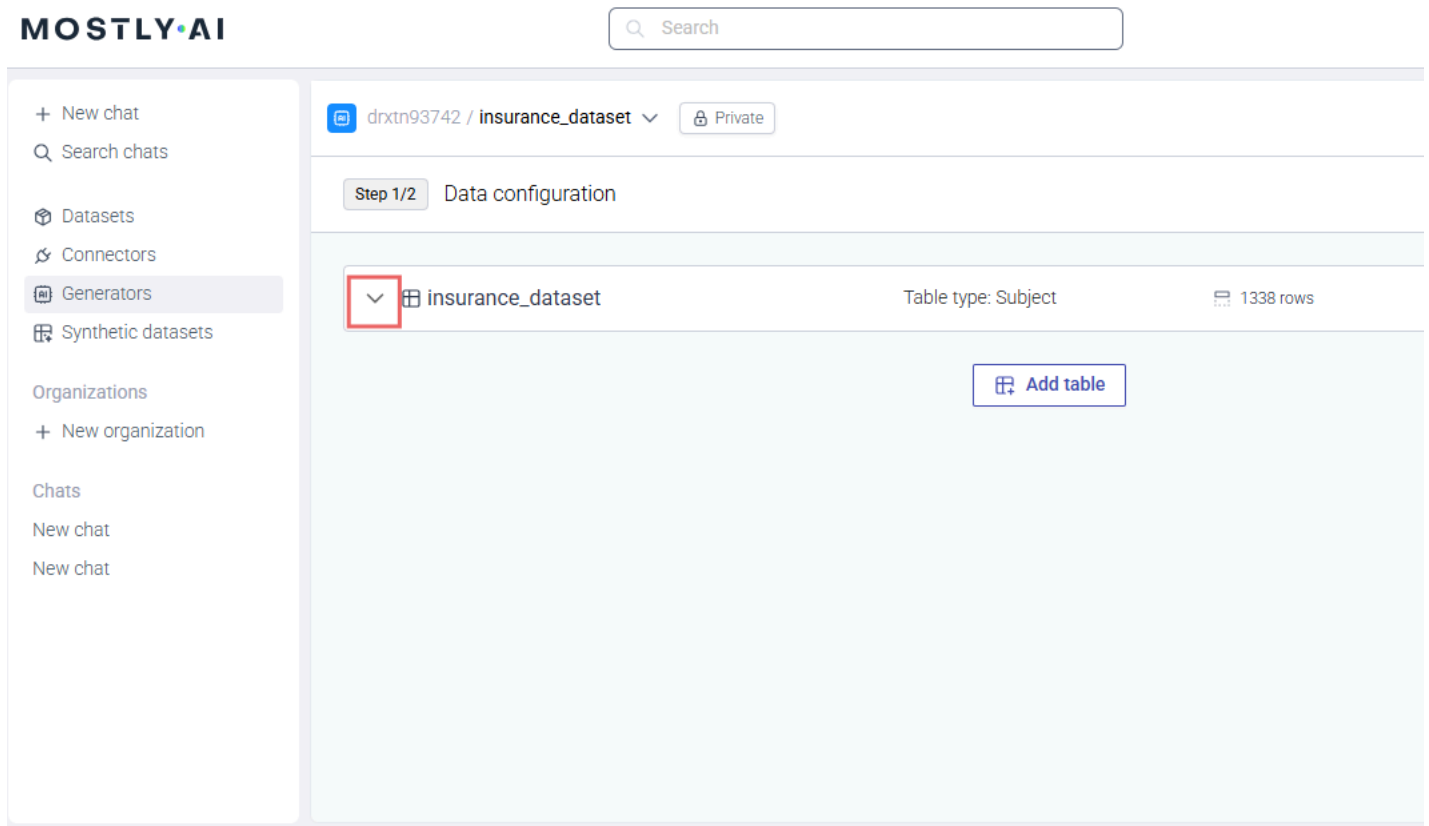
Name	Visibility	Status	Activity
drxtn93742/insurance_dataset (1)	Private	Ready	0 1

- After you upload the data set, you will see its filename on the console. Then select Proceed as seen in the following screen captures:



5. Data configuration settings

- You can choose to modify the category of an attribute, or you can choose to include a parameter in the augmentation process without these settings. For the purposes of this lab, do not change these settings.



- Simply select **Configure models** to go to the model configuration settings.

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Step 1/2

Data configuration

insurance_dataset

Table type: Subject

1338 rows

Table relationships

Table columns

Primary key

Include	Name	Encoding type
☒	age	Tabular/Numeric: Auto
☒	gender	Tabular/Categorical
☒	bmi	Tabular/Numeric: Auto
☒	children	Tabular/Numeric: Auto
☒	smoker	Tabular/Categorical
☒	region	Tabular/Categorical
☒	expenses	Tabular/Numeric: Auto

6. Model configuration settings

You can modify the max training time, number of epochs, sample size, and other settings to generate the best possible model based on your requirements. For the purpose of this lab, use the default settings.

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Step 2/2

Model configuration

Presets

Accuracy

insurance_dataset

tabular

Subject table

1,338 rows

10 min

When you complete working with the settings, select **Start training**. You will find this option on the top right corner of the web page.

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Step 2/2

Model configuration

Presets

Accuracy

insurance_dataset

tabular

Subject table

1,338 rows

10 min

Model

Select the tabular model size.

MOSTLY_AI/Medium

Compute

Select the compute type to train this model.

CPU Intel Xeon Spot: 14 CPUs, 26GB

Training parameters

Adjust the training parameters to prioritize speed over accuracy, or vice versa.

Max training time

10

mins

Max sample size

1,338

7. Model training

After the model training completes, you will see an onscreen result similar to what you see on the following screen capture.

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Private

insurance_dataset

0

0

Created by dxtn93742 • 6 minutes ago

Explore

Description

Add description...

Accuracy

88.3%

Number of tables

1

Data insights

insurance_dataset

tabular

Model report

Sample size

1,338 | 1,338

Accuracy

88.3% | 91.1%

Cosine similarity

0.98208 | 0.99518

Discriminator AUC

58.8% | 45.2%

Distances

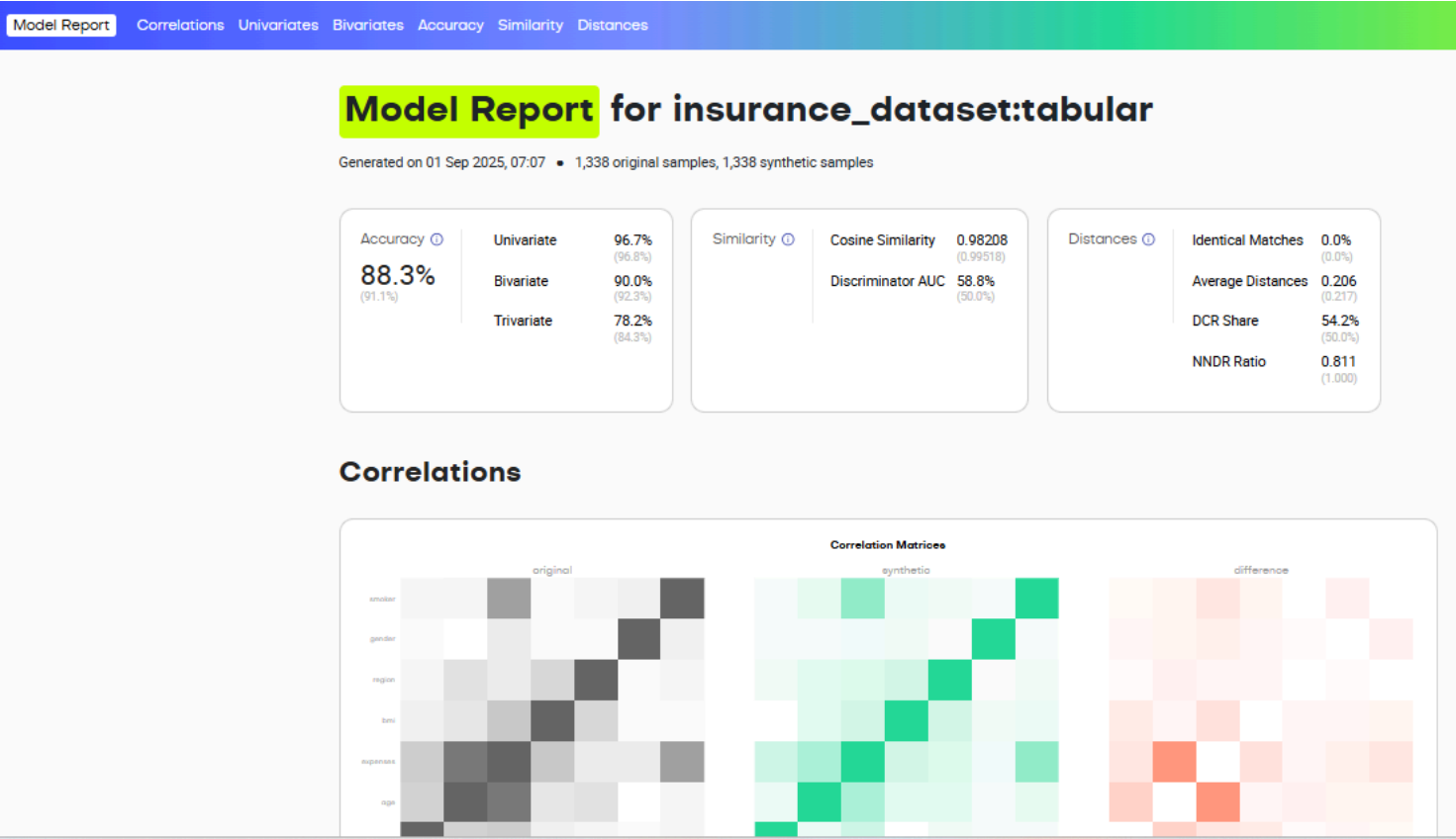
0.206 | 0.217

Model samples

insurance_dataset

age	gender	bmi	children	smoker	region	expenses
19	male	34.9	2	yes	northwest	40302.47
29	female	27.6	0	no	southwest	1356.25

Click the **Model report** hyperlink to open the Quality Assurance Report in a separate tab. The page displays similar to what you see in the following screen capture.



Correlations

original

Correlation Matrices

synthetic

difference

Note that the training accuracy can be different every time the model is trained.

On the original page, click Generate data to use this trained model to generate the required synthetic data.

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0

0

Created by drxtn93742 • 6 minutes ago

Description

Add description...

Accuracy ⓘ

88.3%

Number of tables

1

Data insights

insurance_dataset

tabular

Model report

Sample size ⓘ

1,338 | 1,338

Accuracy ⓘ

88.3% | 91.1%

Cosine similarity ⓘ

0.98208 | 0.99518

Discriminator AUC ⓘ

58.8% | 45.2%

8. Create Synthetic data

You can select the number of samples you want to generate, as well as modify the statistical nature of the data created by choosing the appropriate parameters. For the purpose of this lab, keep all the settings at their default values, and select Start generation to create the required synthetic data.

about:blank

7/9

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Synthetic dataset configuration

Generator used insurance_dataset

^

insurance_dataset

Table type: Subject

1,338 rows

Sample size

Define the number of rows to generate for your synthetic data.

1,338

rows

Conditional simulation

Sampling controls

Fairness

9. Download the synthetic data

After the synthetic data generation is complete, you will see a web page as shown within the following screen capture.

+

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0

0

Created by dxrtn93742 • 1 minute ago

Explore

Download

Description

Add description...

Generator used

insurance_dataset

Number of tables

1

Total generated rows

1,338

Data insights

insurance_dataset

tabular

Model report

Data report

Temperature

1.0

Fairness

Not applied

TopP

1.0

Generate with seed

Not applied

Rebalancing

Not applied

Imputation

Not applied

Generated rows

1,338 | 1,338

Data samples

insurance_dataset

Click on Download synthetic data to download the dataset created. The dataset can be downloaded in any of the available formats.

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insurance_dataset

0

0

Created by drxtn93742 • 1 minute ago

Description

Add description...

Generator used

insurance_dataset

Number of tables

1

Total generated rows

1,338

Data insights

insurance_dataset

tabular

Model report

Data report

Temperature ①	TopP ①	Rebalancing ①	Imputation ①
1.0	1.0	Not applied	Not applied
Fairness ①	Generate with seed ①		
Not applied	Not applied		

Data samples

insurance_dataset

You can now use this synthetic data set for data science operations; or, you can also augment the original data set with these samples.

Conclusion

Congratulations! You have completed the lab on data augmentation using the Mostly.ai tool.

Author(s)

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